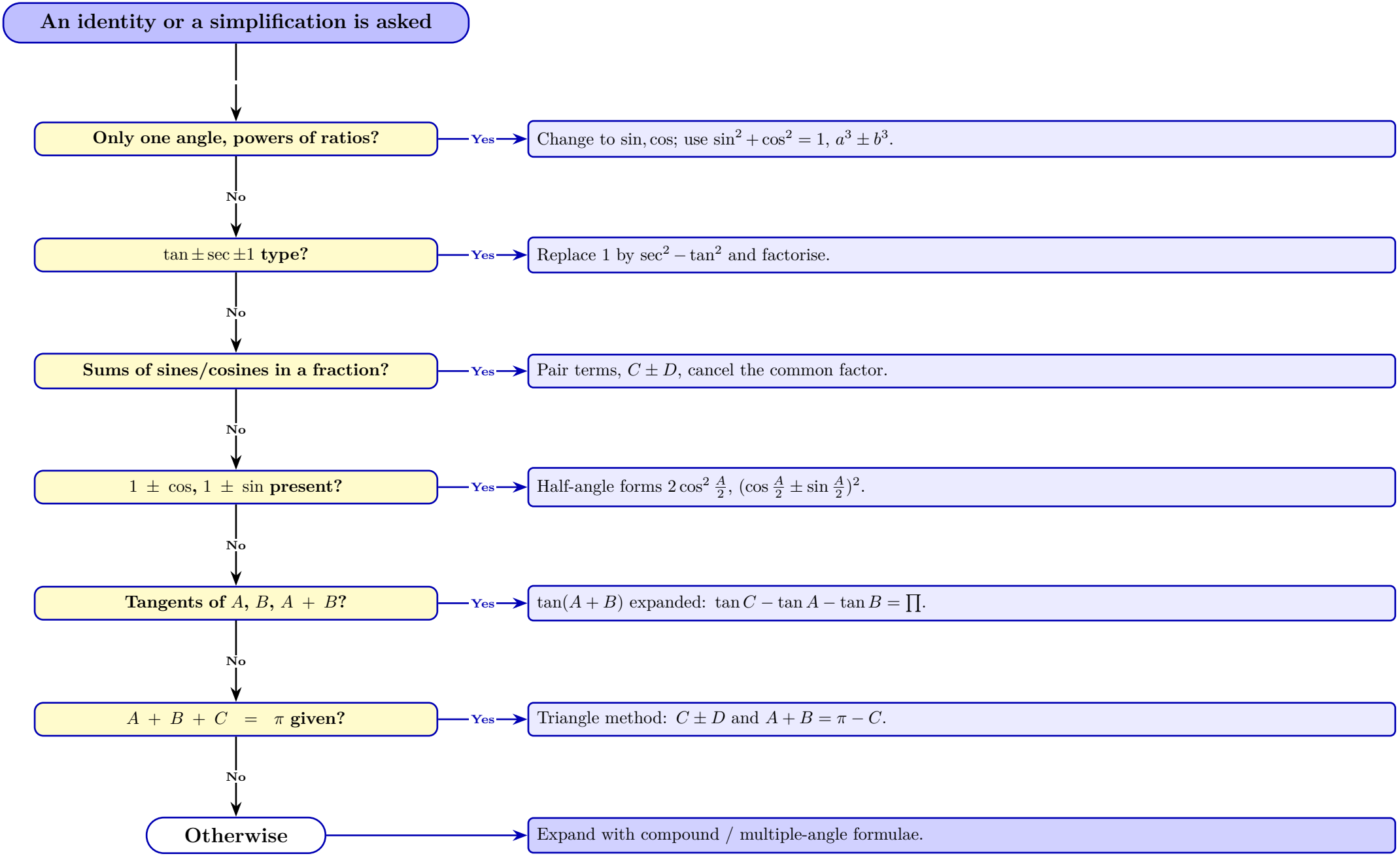
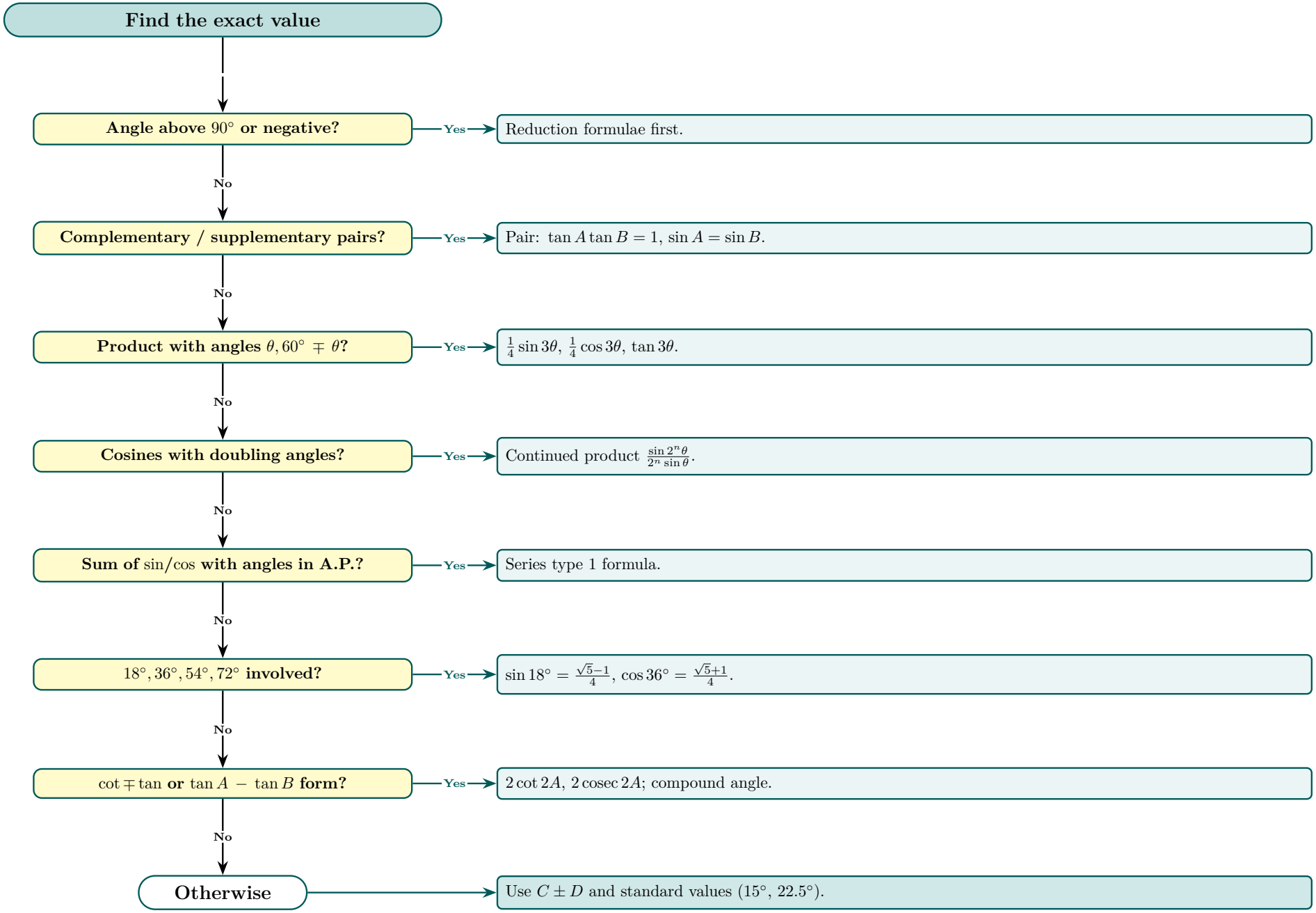
Flowchart 1: Proving an identity



Flowchart 2: Value of an expression in degrees



Flowchart 3: One ratio given, others asked

A value like $\sin A = k$ is given

Ratios of the same angle asked? — Yes → Right triangle + Pythagoras + quadrant sign.

No

Ratios of $2A$ or $3A$ asked? — Yes → $2A$ / $3A$ formulae (in tan if $\tan A$ known).

No

Ratios of $\frac{A}{2}$ asked? — Yes → $\sqrt{\frac{1 \mp \cos A}{2}}$, sign from quadrant of $\frac{A}{2}$.

No

Two angles given, $A \pm B$ asked? — Yes → Find each ratio, then compound angle formula.

No

Ratio $\frac{\sin}{\sin}$ given, $\frac{\tan}{\tan}$ asked? — Yes → Componendo-dividendo, then $C \pm D$.

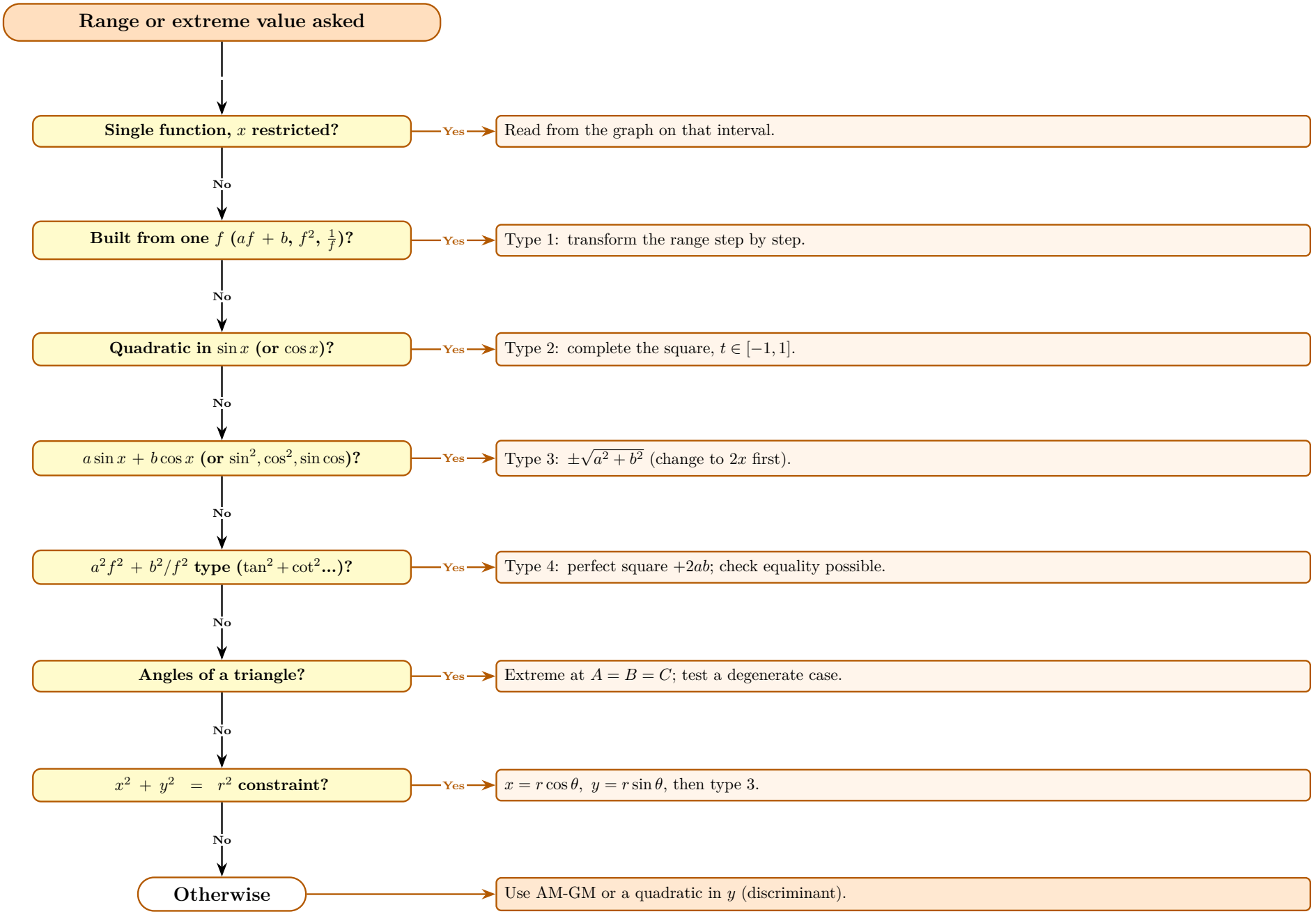
No

$\frac{\sin^4}{a} + \frac{\cos^4}{b} = \frac{1}{a+b}$? — Yes → $\sin^2\theta = \frac{a}{a+b}$.

No

Otherwise → Square and add / use $\sec \pm \tan$ reciprocal pair.

Flowchart 4: Range / maximum-minimum



Flowchart 5: Sum or product of a series

